RAG-Based Customer Support Assistant

High Level Design (HLD)

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📘 1. System Overview

◈ Problem Definition

Customer support systems face challenges in handling repetitive queries, ensuring consistent responses, and scaling efficiently.

This system uses **RAG (Retrieval-Augmented Generation)** to:

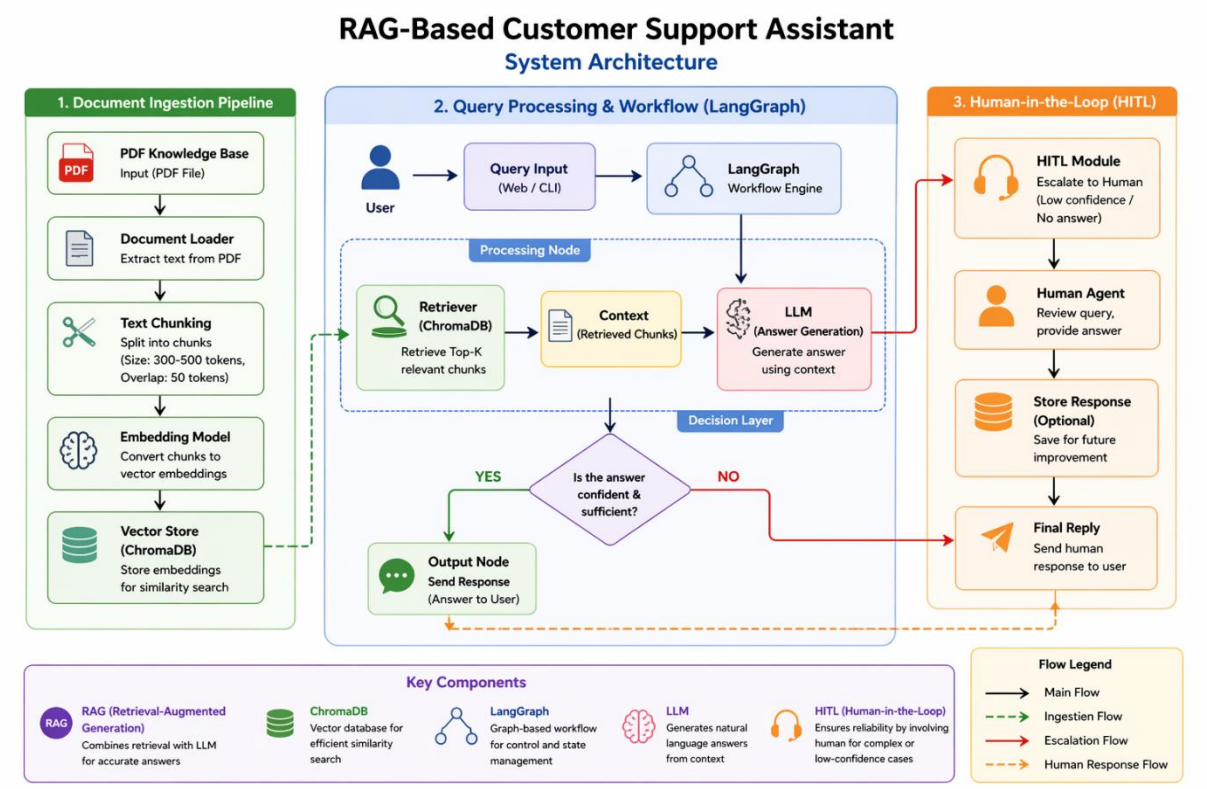
- Retrieve information from a PDF knowledge base
- Generate accurate answers
- Escalate complex queries using HITL

◈ Scope

- PDF-based knowledge system
- Real-time query answering
- Human escalation support
- Graph-based workflow (LangGraph)

📘 2. Architecture Diagram

“System Architecture of RAG-Based Customer Support Assistant”



📘 3. Component Description

◈ Document Loader

Extracts text from PDF using PyPDFLoader

◈ Chunking

Splits text into chunks (300–500 tokens, 50 overlap)

◈ Embedding Model

Converts text into vector representations

◈ Vector Store (ChromaDB)

Stores embeddings for similarity search

◈ Retriever

Fetches top relevant chunks (Top-K = 3–5)

◈ LLM

Generates answers using retrieved context

◈ LangGraph

Controls workflow using graph-based execution

◈ Routing Layer

Decides whether to:

- Answer directly
- Escalate to human

◈ HITL Module

Handles low-confidence or complex queries

📘 4. Data Flow

📥 Document Flow

PDF → Loader → Chunking → Embedding → ChromaDB

📤 Query Flow

User Query → Retrieve → LLM → Decision → Response / HITL

📗 5. Technology Choices

Component	Technology	Reason
Vector DB	ChromaDB	Fast and lightweight
Workflow	LangGraph	State-based control
LLM	OpenAI	High Accuracy
UI	Streamlit	Easy to Build

📓 6. Scalability Considerations

- Efficient chunking for large documents
- Caching frequent queries
- Optimized retrieval (Top-K tuning)
- Scalable vector database
- Async processing to reduce latency